9.3.1 Fisher's Ratio

Q: $R = \frac{(m_1 - m_2)^2}{\sigma_1^2 + \sigma_2^2}$

C_1	1.2	1	0.9	1.1
C_2	2.2	2.3	2.1	2

$R = ?$

Solution ① $m_1 = \frac{1}{4} \times (1.2 + 1 + 0.9 + 1.1)$

$= \frac{1}{4} \times 4.2$

$= 1.05$

② $m_2 = \frac{1}{4} \times (2.2 + 2.3 + 2.1 + 2)$

$= 2.15$

③ $\sigma_1^2 = \frac{1}{4} \sum_{k=1}^{4} (X_{1k} - m_1)^2$

$= \frac{1}{4} \times [0.0225 + 0.0025 + 0.0225 + 0.0025]$

$= 0.0125$

④ $\sigma_2^2 = \frac{1}{4} \times 0.05$

$= 0.0125$

⑤ $R = \frac{(1.05 - 2.15)^2}{0.0125 + 0.0125} = 48.4$